Day 12 Study Notes

Tue 24 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Ray optics (part 1) (75 min) · Maths: Binomial theorem (part 1) (75 min) · Chemistry: General organic chemistry (part 3: isomerism) (60 min)
Exam	1 h 15 min	The Day 12 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Ray optics (part 1)

Ray optics gives two questions in most papers. Today covers mirrors and refraction at flat surfaces. Part 2 covers lenses and part 3 prisms and instruments.

1. Sign convention (Cartesian)

- Measure all distances from the pole of the mirror (or the optical centre of the lens).
- Distances in the direction of the incident light are positive; against it, negative. With light coming from the left, a real object always has negative u.
- Heights above the principal axis are positive, below negative.
- For a concave mirror f is negative; for a convex mirror f is positive.

2. Plane mirrors

- The image is virtual, erect, the same size, and as far behind the mirror as the object is in front.
- If the mirror turns through θ, the reflected ray turns through 2θ.
- Two mirrors at angle θ: number of images $= \dfrac{360^\circ}{\theta} - 1$ when $\dfrac{360^\circ}{\theta}$ is even (and for an object on the bisector when it is odd).
- To see your full height you need a mirror at least half your height.

3. Spherical mirrors

- $f = \dfrac{R}{2}$. Mirror formula: $\dfrac{1}{v} + \dfrac{1}{u} = \dfrac{1}{f}$. Magnification: $m = -\dfrac{v}{u} = \dfrac{h_i}{h_o}$.
- Negative m: real and inverted. Positive m: virtual and erect. $|m| > 1$: magnified.

Object (concave mirror)	Image	Nature
At infinity	At F	Real, inverted, point-sized
Beyond C	Between F and C	Real, inverted, diminished
At C	At C	Real, inverted, same size
Between C and F	Beyond C	Real, inverted, magnified
At F	At infinity	Real, inverted, very large
Between F and P	Behind the mirror	Virtual, erect, magnified

- A **convex mirror** always gives a virtual, erect, diminished image between P and F. That is why it is used as a rear-view mirror: it shows a wide field.

4. Refraction at a flat surface

- Refractive index $n = \dfrac{c}{v}$. Snell's law: $n_1 \sin i = n_2 \sin r$.
- Going into a denser medium the ray bends towards the normal; speed and wavelength fall, frequency stays the same.
- **Apparent depth:** looking straight down from air into a medium of index n, an object at real depth d appears at $\dfrac{d}{n}$. It appears raised by $d\left(1 - \dfrac{1}{n}\right)$.

- **Glass slab:** an object seen through a slab of thickness t appears shifted towards the slab by $t\left(1-\frac{1}{n}\right)$. The emerging ray is parallel to the incident ray, displaced sideways.

5. Total internal reflection

- Happens only when light goes from a denser to a rarer medium and the angle of incidence exceeds the critical angle C.
- $\sin C = \frac{1}{n}$ for a medium against air (more generally $\frac{n_{rarer}}{n_{denser}}$). For glass ($n = 1.5$), $C \approx 42^\circ$; for $n = \sqrt{2}$, $C = 45^\circ$.
- Uses and examples: optical fibres, sparkle of diamond, totally reflecting prisms, mirage.

> **Solved example 1.** An object is 30 cm in front of a concave mirror of focal length 20 cm. Find the image.
> $u = -30$, $f = -20$: $\frac{1}{v} = -\frac{1}{20} + \frac{1}{30} = -\frac{1}{60}$, so $v = -60$ cm. $m = -\frac{v}{u} = -2$: real, inverted, twice the size, 60 cm in front.

> **Solved example 2.** A coin lies at the bottom of a tank of water ($n = \frac{4}{3}$) 12 cm deep. How deep does it appear?
> $\frac{12}{4/3} = 9$ cm. It appears raised by 3 cm.

> **Traps.** Put the signs into u, v and f before solving, and read the sign of v at the end. A mirage is total internal reflection, not dispersion. Frequency does not change on refraction. Total internal reflection cannot happen going from air into glass.

Mathematics · Binomial theorem (part 1)

The binomial theorem gives one question in nearly every paper, usually a general-term problem: a particular coefficient or the term independent of x. Today covers the theorem and the general term. Part 2 covers the properties of the coefficients and remainder problems.

1. The theorem

- $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$ for a positive integer n, where $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$.
- There are $n+1$ terms. The powers of a fall from n to 0 while the powers of b rise from 0 to n, and in every term they add up to n.
- $\binom{n}{r} = \binom{n}{n-r}$, so coefficients equally far from the two ends are equal.
- Pascal's rule: $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$.

2. The general term

- $T_{r+1} = \binom{n}{r} a^{n-r} b^r$. The $(r+1)$th term carries $\binom{n}{r}$, so the 4th term has $r = 3$.
- For $(a-b)^n$, use $b \to -b$: the term carries $(-1)^r$.
- The rth term from the end of $(a+b)^n$ is the $(n-r+2)$th term from the start.

3. Method for a coefficient or the term independent of x

1. Write T_{r+1} with all the powers of x collected: $x^{(\text{something in } r)}$.
2. Set that power equal to the power you want (0 for the term independent of x) and solve for r.
3. If r is not a whole number between 0 and n, there is no such term.
4. Put r back to get the coefficient, keeping the signs and the numerical factors.

4. Middle terms

- n even: one middle term, the $\left(\dfrac{n}{2}+1\right)$th.
- n odd: two middle terms, the $\left(\dfrac{n+1}{2}\right)$th and $\left(\dfrac{n+3}{2}\right)$th.
- The middle term has the greatest binomial coefficient.

5. Approximation

- For small x: $(1+x)^n \approx 1 + nx$. Example: $(1.01)^{10} \approx 1.1$; the true value is about 1.1046, slightly more.

Solved example 1. Find the term independent of x in $\left(x^2 + \dfrac{1}{x}\right)^9$.
$T_{r+1} = \binom{9}{r} x^{2(9-r)} x^{-r} = \binom{9}{r} x^{18-3r}$. $18 - 3r = 0$ gives $r = 6$. The term is $\binom{9}{6} = 84$.

Solved example 2. Find the 4th term of $(2x-3)^5$.
$r = 3$: $T_4 = \binom{5}{3}(2x)^2(-3)^3 = 10 \times 4x^2 \times (-27) = -1080x^2$.

Traps. The $(r+1)$th term uses $\binom{n}{r}$; an off-by-one here is the most common mistake. Keep the numerical factors inside the brackets, such as 2^{n-r} in $(2x+\dots)^n$. Keep the sign $(-1)^r$ for $(a-b)^n$. A coefficient and a term are different: the term includes the power of x.

Chemistry · General organic chemistry (part 3: isomerism)

Isomerism gives one question in most papers, usually counting isomers or picking the compound that shows a given kind of isomerism. Learn the conditions for each kind, then practise counting.

1. Kinds of isomerism

Isomers have the same molecular formula but different arrangements. **Structural isomers** differ in which atoms are bonded to which. **Stereoisomers** have the same bonds but a different arrangement in space.

2. Structural isomerism

Kind	What differs	Example
Chain	The carbon skeleton	Butane and 2-methylpropane
Position	Position of a group or multiple bond	Propan-1-ol and propan-2-ol; but-1-ene and but-2-ene
Functional	The functional group	Ethanol and dimethyl ether; propanal and propanone
Metamerism	Alkyl groups on either side of the same functional group	Ethoxyethane and methoxypropane
Tautomerism	Position of a hydrogen, the two forms in equilibrium	Keto and enol forms of acetone
Ring–chain	Open chain against ring	Propene and cyclopropane

- **Keto–enol tautomerism** needs at least one hydrogen on the carbon next to the C=O (an α hydrogen). Benzaldehyde, formaldehyde and $(CH_3)_3CCHO$ have none, so they do not show it.
- Useful counts: C_4H_{10} 2 isomers, C_5H_{12} 3, C_6H_{14} 5. $C_4H_{10}O$: 4 alcohols and 3 ethers. C_3H_6O: propanal, propanone and others.

3. Geometrical (cis–trans) isomerism

- Needs restricted rotation (a C=C double bond or a ring) **and** two different groups on each carbon of the double bond.
- But-2-ene shows it. But-1-ene and 2-methylpropene do not, because one carbon carries two identical H atoms or two identical CH_3 groups.
- Cis isomers usually have higher dipole moment and boiling point; trans isomers are usually more stable and pack better, so they have higher melting points.
- When the groups are all different, the E/Z system is used: rank the groups on each carbon by atomic number; Z if the higher-ranked groups are on the same side.

4. Optical isomerism

- A molecule is **chiral** if it is not superimposable on its mirror image. The common cause is a **chiral centre**: a carbon carrying four different groups.
- **Enantiomers** are mirror images. They have identical physical properties except the direction in which they rotate plane-polarised light.
- **Diastereomers** are stereoisomers that are not mirror images. They have different physical properties.

- A **racemic mixture** has equal amounts of two enantiomers and is optically inactive (external compensation).
- A **meso compound** has chiral centres but an internal plane of symmetry, so it is optically inactive (internal compensation). Example: meso-tartaric acid.
- With n different chiral centres and no symmetry, there are 2^n stereoisomers. Symmetric molecules have fewer, because some are meso.
- A molecule with both a cis–trans double bond and a chiral centre has $2 \times 2 = 4$ stereoisomers.

> **Solved example 1.** Which of butan-1-ol and butan-2-ol is chiral?
> In butan-2-ol, C-2 carries H, OH, CH_3 and C_2H_5: four different groups. So butan-2-ol is chiral. Butan-1-ol has no such carbon.

> **Solved example 2.** How many stereoisomers does $CH_3CH(Cl)CH(Br)CH_3$ have?
> Two different chiral centres and no symmetry: $2^2 = 4$.

Traps. A double bond with two identical groups on one carbon shows no cis–trans isomerism. Enantiomers have the same boiling and melting points. A racemic mixture and a meso compound are both inactive, for different reasons. Count the C–H hydrogens when checking for α hydrogens.